PSEUDOCODE & JAVA CONVERSION

Question 1: Check Whether a Number is Positive or Negative

- Pseudocode

```
BEGIN
        DECLARE num AS INTEGER
        READ num
        IF num >= 0 THEN
          PRINT "Positive Number"
        ELSE
          PRINT "Negative Number"
        ENDIF
       END
```

- Java Code

```
import java.util.Scanner;

public class PositiveNegative {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int num;

    System.out.print("Enter a number: ");
    num = sc.nextInt();

    if (num >= 0) {
```

```
            System.out.println("Positive Number");
        } else {
            System.out.println("Negative Number");
        }
    }
}
```

Question 2: Check Even or Odd Number.

- Pseudocode

```
BEGIN
  DECLARE num AS INTEGER
  READ num

  IF num % 2 == 0 THEN
    PRINT "Even Number"
  ELSE
    PRINT "Odd Number"
  ENDIF
END
```

- Java Code

```
import java.util.Scanner;

public class EvenOdd {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
```

```
    int num;

    System.out.print("Enter a number: ");
    num = sc.nextInt();

    if (num % 2 == 0) {
      System.out.println("Even Number");
    } else {
      System.out.println("Odd Number");
    }
  }
}
```

Question 3: Find Largest Among Two Numbers.

- Pseudocode

```
BEGIN
  DECLARE a, b AS INTEGER
  READ a, b

  IF a > b THEN
    PRINT "A is Largest"
  ELSE
    PRINT "B is Largest"
  ENDIF
END
```

- Java Code

```java
import java.util.Scanner;

public class LargestTwo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int a, b;

        System.out.print("Enter first number: ");
        a = sc.nextInt();

        System.out.print("Enter second number: ");
        b = sc.nextInt();

        if (a > b) {
            System.out.println("A is Largest");
        } else {
            System.out.println("B is Largest");
        }
    }
}
```

Question 4: Grade Calculation Based on Marks

- Pseudocode

```
BEGIN
  DECLARE marks AS INTEGER
  READ marks
```

```
IF marks >= 80 THEN
   PRINT "Grade A"
ELSE IF marks >= 60 THEN
   PRINT "Grade B"
ELSE IF marks >= 50 THEN
   PRINT "Grade C"
ELSE IF marks >= 40 THEN
   PRINT "Grade D"
ELSE
   PRINT "Fail"
ENDIF
END
```

- Java Code

```
import java.util.Scanner;

public class GradeSystem {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);
      int marks;

      System.out.print("Enter marks: ");
      marks = sc.nextInt();

      if (marks >= 80) {
```

```
        System.out.println("Grade A");
      } else if (marks >= 60) {
        System.out.println("Grade B");
      } else if (marks >= 50) {
        System.out.println("Grade C");
      } else if (marks >= 40) {
        System.out.println("Grade D");
      } else {
        System.out.println("Fail");
      }
    }
}
```

Question 5: Addition of Two Numbers

- Pseudocode

```
BEGIN
  DECLARE a, b, sum AS INTEGER
  READ a, b
  sum = a + b
  PRINT sum
END
```

- Java Code

```
import java.util.Scanner;

public class Addition {
  public static void main(String[] args) {
```

```
    Scanner sc = new Scanner(System.in);
    int a, b, sum;

    System.out.print("Enter first number: ");
    a = sc.nextInt();

    System.out.print("Enter second number: ");
    b = sc.nextInt();

    sum = a + b;

    System.out.println("Sum = " + sum);
  }
}
```

Question 6: Print Numbers from 1 to 10 (For Loop)

- Pseudocode

```
BEGIN
  DECLARE num AS INTEGER
  READ num

  IF num >= 0 THEN
    PRINT "Positive Number"
  ELSE
    PRINT "Negative Number"
  ENDIF
END
```

- Java Code

```
import java.util.Scanner;


public class PositiveNegative {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int num;


    System.out.print("Enter a number: ");
    num = sc.nextInt();


    if (num >= 0) {
      System.out.println("Positive Number");
    } else {
      System.out.println("Negative Number");
    }
  }
}
```

Question 7: Print Numbers 10 to 1 (Reverse Order)

- Pseudocode

```
BEGIN
  FOR i = 10 TO 1 STEP -1
    PRINT i
  END FOR
END
```

- Java Code

```
public class ReversePrint {
   public static void main(String[] args) {

      for (int i = 10; i >= 1; i--) {
         System.out.println(i);
      }

   }
}
```

Question 8: Sum of First N Natural Numbers

- Pseudocode

```
BEGIN
  DECLARE n, sum AS INTEGER
  sum = 0
  READ n

  FOR i = 1 TO n
     sum = sum + i
  END FOR

  PRINT sum
END
```

- Java Code

```
import java.util.Scanner;

public class SumOfN {
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, sum = 0;

        System.out.print("Enter value of n: ");
        n = sc.nextInt();

        for (int i = 1; i <= n; i++) {
            sum = sum + i;
        }

        System.out.println("Sum = " + sum);
    }
}
```

Question 9: While Loop – Print 1 to 10

- Pseudocode

```
BEGIN
  SET i = 1
  WHILE i <= 10
    PRINT i
    i = i + 1
  END WHILE
END
```

- Java Code

```
public class WhileExample {
```

```
  public static void main(String[] args) {

    int i = 1;

    while (i <= 10) {
      System.out.println(i);
      i++;
    }

  }
}
```

Question 10: Largest Among Three Numbers

- Pseudocode

```
BEGIN
  DECLARE a, b, c AS INTEGER
  READ a, b, c

  IF a > b AND a > c THEN
    PRINT "A is Largest"
  ELSE IF b > a AND b > c THEN
    PRINT "B is Largest"
  ELSE
    PRINT "C is Largest"
  ENDIF
END
```

- Java Code

```
import java.util.Scanner;

public class LargestThree {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      int a, b, c;

      System.out.print("Enter three numbers: ");
      a = sc.nextInt();
      b = sc.nextInt();
      c = sc.nextInt();

      if (a > b && a > c) {
         System.out.println("A is Largest");
      } else if (b > a && b > c) {
         System.out.println("B is Largest");
      } else {
         System.out.println("C is Largest");
      }
   }
}
```

Question 11: Check a > b AND b > c

- Pseudocode

```
BEGIN
  DECLARE a, b, c AS INTEGER
```

```
READ a, b, c

IF a > b AND b > c THEN
  PRINT "Condition True"
ELSE
  PRINT "Condition False"
ENDIF
END
```

- Java Code

```
import java.util.Scanner;

public class LogicalCheck {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int a, b, c;

    System.out.print("Enter three numbers: ");
    a = sc.nextInt();
    b = sc.nextInt();
    c = sc.nextInt();

    if (a > b && b > c) {
      System.out.println("Condition True");
    } else {
      System.out.println("Condition False");
```

```
    }
  }
}
```

Question 12: Switch Case Example

- Pseudocode

```
BEGIN
 DECLARE x AS INTEGER
 READ x

 SWITCH x
   CASE 1: PRINT "ONE"
   CASE 2: PRINT "TWO"
   CASE 3: PRINT "THREE"
   DEFAULT: PRINT "WRONG"
 END SWITCH
END
```

- Java Code

```
import java.util.Scanner;

public class SwitchExample {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int x;
    System.out.print("Enter a number (1-3): ");
    x = sc.nextInt();
```

```
    switch (x) {
      case 1:
        System.out.println("ONE");
        break;
      case 2:
        System.out.println("TWO");
        break;
      case 3:
        System.out.println("THREE");
        break;
      default:
        System.out.println("WRONG");
    }
  }
}
```

Question 13: Print Multiplication Table of a Number

- Pseudocode

```
BEGIN
  DECLARE num AS INTEGER
  READ num

  FOR i = 1 TO 10
    PRINT num * i
  END FOR
END
```

- Java Code

```
import java.util.Scanner;


public class MultiplicationTable {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      int num;
      System.out.print("Enter number: ");
      num = sc.nextInt();


      for (int i = 1; i <= 10; i++) {
         System.out.println(num + " x " + i + " = " + (num * i));
      }
   }
}
```

Question 14: Print Numbers from 1 to 10 (Step 2)

- Pseudocode

```
BEGIN
  FOR i = 1 TO 10 STEP 2
    PRINT i
  END FOR
END
```

- Java Code

```
public class StepTwo {
   public static void main(String[] args) {
```

```
    for (int i = 1; i <= 10; i += 2) {
      System.out.println(i);
    }

  }
}
```

Question 15: Multiply Two Numbers Using Loop

- Pseudocode

```
BEGIN
  DECLARE a, b, result AS INTEGER
  result = 0
  READ a, b

  FOR i = 1 TO b
    result = result + a
  END FOR

  PRINT result
END
```

- Java Code

```
import java.util.Scanner;

public class MultiplyUsingLoop {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
```

```
    int a, b, result = 0;

    System.out.print("Enter two numbers: ");
    a = sc.nextInt();
    b = sc.nextInt();

    for (int i = 1; i <= b; i++) {
      result = result + a;
    }

    System.out.println("Result = " + result);
  }
}
```

Question 16: Armstrong Number

- Pseudocode

```
BEGIN
DECLARE n, original, digit, sum AS INTEGER
SET sum = 0
READ n
original = n
WHILE n > 0
        digit = n MOD 10
        sum = sum + digit * digit * digit
```

```
        n = n / 10

        END WHILE

IF sum = original THEN

        PRINT "Armstrong"

        ELSE PRINT "Not Armstrong"

ENDIF

END
```

- Java Code

```
import java.util.Scanner;

public class ArmstrongNumber {

        public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

       int n, original, digit, sum = 0;
      System.out.print("Enter number: ");
      n = sc.nextInt();
      original = n;
      while (n > 0) {
         digit = n % 10;
         sum = sum + (digit * digit * digit);
         n = n / 10;
      }

     if (sum == original) {
         System.out.println("Armstrong");
     } else {
         System.out.println("Not Armstrong");
     }
}

}
```

Question 17: Check Palindrome Number

- Pseudocode

```
BEGIN
 DECLARE n, original, digit, reverse AS INTEGER
 SET reverse = 0
 READ n
 original = n

 WHILE n > 0
   digit = n MOD 10
   reverse = reverse * 10 + digit
   n = n / 10
 END WHILE

 IF original = reverse THEN
   PRINT "Palindrome"
 ELSE
   PRINT "Not Palindrome"
 ENDIF
END
```

- Java Code

```
import java.util.Scanner;
```

```
public class PalindromeNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n, original, digit, reverse = 0;

        System.out.print("Enter number: ");
        n = sc.nextInt();
        original = n;

        while (n > 0) {
            digit = n % 10;
            reverse = reverse * 10 + digit;
            n = n / 10;
        }

        if (original == reverse) {
            System.out.println("Palindrome");
        } else {
            System.out.println("Not Palindrome");
        }
```

```
  }
}
```

Question 18: Fibonacci Series

- Pseudocode

```
BEGIN
 DECLARE n, a, b, c, i AS INTEGER
 SET a = 0
 SET b = 1
 READ n

 PRINT a
 PRINT b

 FOR i = 2 TO n
   c = a + b
   PRINT c
   a = b
   b = c
 END FOR
END
```

- Java Code

```
import java.util.Scanner;
```

```
public class FibonacciSeries {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int n, a = 0, b = 1, c;

    System.out.print("Enter number of terms: ");
    n = sc.nextInt();

    System.out.print(a + " " + b + " ");

    for (int i = 2; i < n; i++) {
      c = a + b;
      System.out.print(c + " ");
      a = b;
      b = c;
    }
  }
}
```

Question 19: Count Number of Digits in a Number

- Pseudocode

```
BEGIN
 DECLARE n, count AS INTEGER
 SET count = 0
 READ n

 WHILE n > 0
   n = n / 10
   count = count + 1
 END WHILE

 PRINT count
END
```

- Java Code

```
import java.util.Scanner;

public class CountDigits {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int n, count = 0;

    System.out.print("Enter a number: ");
```

```
        n = sc.nextInt();


        while (n > 0) {
            n = n / 10;
            count++;
        }


        System.out.println("Number of digits = " + count);
    }
}
```

Question 20: Separate Two Digit Number (Tens & Ones)

- Pseudocode

```
BEGIN
  DECLARE n, a, b AS INTEGER
  READ n


  a = n MOD 10
  b = n / 10


  PRINT b
  PRINT a
END
```

- Java Code

```
import java.util.Scanner;


public class SeparateDigits {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);


      int n, a, b;


      System.out.print("Enter two digit number: ");
      n = sc.nextInt();


      a = n % 10;
      b = n / 10;


      System.out.println("Tens = " + b);
      System.out.println("Ones = " + a);
   }
}
```

Question 21: Reverse a Number

- Pseudocode

BEGIN

```
 DECLARE n, digit, reverse AS INTEGER
 SET reverse = 0
 READ n

 WHILE n > 0
   digit = n MOD 10
   reverse = reverse * 10 + digit
   n = n / 10
 END WHILE

 PRINT reverse
END
```

- Java Code

```
import java.util.Scanner;

public class ReverseNumber {
  public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    int n, digit, reverse = 0;

    System.out.print("Enter number: ");
```

```
        n = sc.nextInt();


        while (n > 0) {
            digit = n % 10;
            reverse = reverse * 10 + digit;
            n = n / 10;
        }


        System.out.println("Reverse = " + reverse);
    }
}
```